

Examine how people use data to develop scientific explanations

Learning intention

- examine how people use data to develop scientific explanations.

Teach and model

- exploring how farmers use soil tests to monitor and manage the health of farms.
- viewing a documentary or webinar and observing how scientists and researchers share their data and explanations.
- exploring age-appropriate science reports and journal articles and identifying where in the text the author has included data, findings or explanations.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.